

Monday, July 6, 2026 1:32 PM

Example 3

Consider two events A and B, where $P(A) = .4$ and $P(B) = .2$.

1. If A and B are independent, what is, $P(A|B)$, $P(A \cap B)$, $P(A \cup B)$?

2. If $P(A|B) = .6$, then are A and B independent? Find $P(A \cap B)$ and $P(B|A)$.

Statistics and Data Analysis – Summer 2026

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1. A and B independent $\Rightarrow P(A|B) = P(A)$.

$$P(A|B) = P(A) = .4$$

$$P(A \cap B) = P(A|B)P(B) = (.4)(.2) = .08$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = .4 + .2 - .08 = .52$$

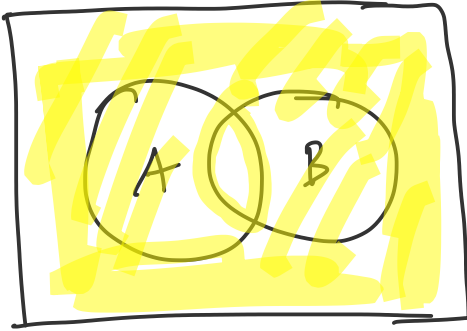
2. $P(A|B) = .6 \neq P(A) \Rightarrow A$ and B are not ind.!

$$P(A \cap B) = P(A|B)P(B) = (.6)(.2) = .12$$

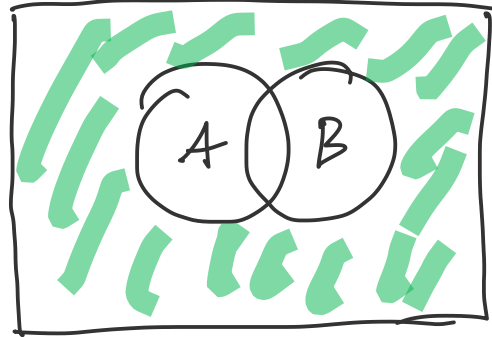
$$P(B|A) = ?$$

$$P(A \cap B) = P(B|A) P(A)$$

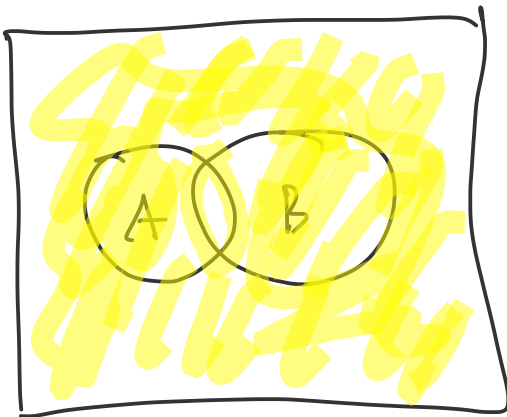
$$.12 = P(B|A) (.4) \Rightarrow P(B|A) = \frac{.12}{.4} = .3$$



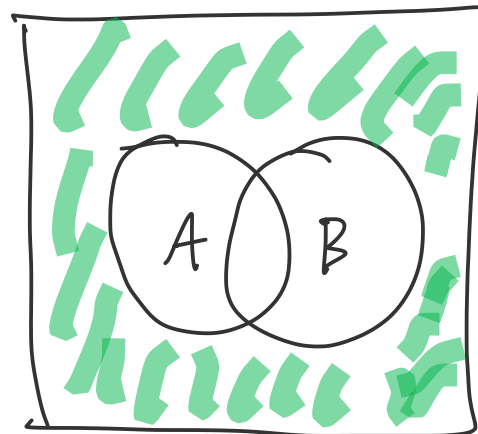
$$P(A \cap \bar{B})$$



$$P(A \cup \bar{B})$$



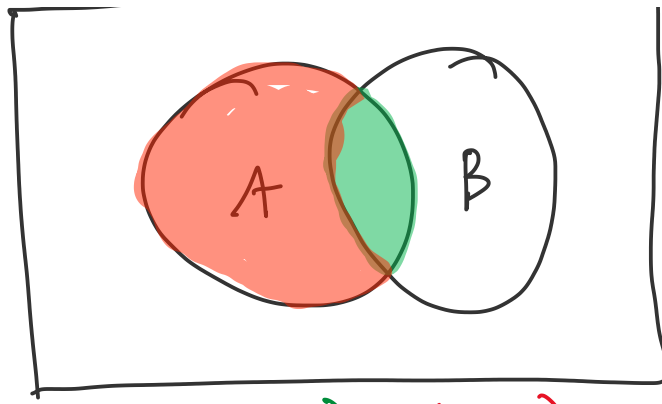
$$P(\bar{A} \cup \bar{B})$$



$$P(\bar{A} \cap \bar{B})$$

Bayes' Theorem

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A|B)P(B)}{P(A)} = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|\bar{B})P(\bar{B})}$$



$$P(A) = P(A \cap B) + P(A \cap \bar{B})$$

$$= P(A|B)P(B) + P(A|\bar{B})P(\bar{B})$$

Example

$$P(\text{Dis}) = .01$$

$$P(\text{Pos}|\text{Dis}) = .98$$

$$P(\text{Neg}|\bar{\text{Dis}}) = .99$$

$$P(\text{Dis}|\text{Pos}) = \frac{P(\text{Pos}|\text{Dis})P(\text{Dis})}{P(\text{Pos}|\text{Dis})P(\text{Dis}) + P(\text{Pos}|\bar{\text{Dis}})P(\bar{\text{Dis}})} = \frac{(.98)(.01)}{(.98)(.01) + (.01)(.99)}$$

$$\frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|\bar{B})P(\bar{B})} = P(B|A)$$

	Dis	$\bar{\text{Dis}}$	
Poc	98	99	197

	10	91	111
Neg	2	9801	9803
	100	9900	10,000

Factorials and Permutations

- In how many ways can we place 4 people on a bench? $4 \cdot 3 \cdot 2 \cdot 1 = 24$
- “x factorial” = $x! = x(x - 1)(x - 2) \dots 1$
- The number of ways we can order x (distinct) elements in a row is $x!$
- By definition: $0! = 1$
- Permutations Example: There are 10 students running for president, vice president, and treasurer of a new club. How many different leadership teams exist?

$$(10)(9)(8) = \frac{10!}{7!} = \frac{10!}{(10-3)!}$$

$$\frac{n!}{(n-k)!} \quad \leftarrow n_p k$$

Number of permutations : $\frac{10!}{(10-3)!}$

Number of combinations : $\frac{10!}{(10-3)!} \div 3! = \frac{10!}{(10-3)! \cdot 3!}$

$$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!} = n_c k$$

Combinations – Example

- An investor is considering investing in 5 firms, A, B, C, D, and E.
 1. If she chose two firms at random, what is the probability that these are A and B?
 2. If she chose three firms at random, what is the probability that E is among them?

$$1. \frac{1}{\binom{5}{2}} = \frac{1}{10} \quad 2. \frac{\binom{4}{2}}{\binom{5}{3}} = \frac{6}{10} = \frac{3}{5}$$

Discrete Random Variables

Let X be a Random Variable (RV)
that represents the face of a fair die

$$P(X=x) = p(x)$$

pdf: probability distribution function

x	$p(x)$
1	$\frac{1}{6}$
2	$\frac{1}{6}$
3	$\frac{1}{6}$
4	$\frac{1}{6}$
5	$\frac{1}{6}$
6	$\frac{1}{6}$

$$\mu_x = E(X) = \sum_{\text{all } x} x p(x)$$

$$= (1)\left(\frac{1}{6}\right) + (2)\left(\frac{1}{6}\right) + (3)\left(\frac{1}{6}\right) + (4)\left(\frac{1}{6}\right) + (5)\left(\frac{1}{6}\right) + (6)\left(\frac{1}{6}\right) = 3\frac{1}{2}$$

$$\sigma_x^2 = \text{Var}(X) = \sum_{\text{all } x} (x - \mu_x)^2 p(x) = \left[\sum_{\text{all } x} x^2 p(x) \right]$$

$$= (1)^2\left(\frac{1}{6}\right) + (2)^2\left(\frac{1}{6}\right) + (3)^2\left(\frac{1}{6}\right) + (4)^2\left(\frac{1}{6}\right) + (5)^2\left(\frac{1}{6}\right) + (6)^2\left(\frac{1}{6}\right) -$$

$$\sigma_x = \sqrt{\text{Var}(X)}$$

Properties of Expected Value

$$\mu_x = E(X)$$

$$E(a) = a$$

$$E(aX) = a\mu_x$$

$$E(X+Y) = \mu_x + \mu_y$$

$$E(aX+bY) = a\mu_x + b\mu_y$$

Properties of Variance

$$\begin{aligned} \sigma_x^2 = \text{Var}(X) &= E[(X - \mu_x)^2] = E[X^2 - 2\mu_x X + \mu_x^2] \\ &= E(X^2) - 2\mu_x E(X) + E(\mu_x^2) \\ &= E(X^2) - 2\mu_x^2 + \mu_x^2 \\ &= E(X^2) - \mu_x^2 \end{aligned}$$

$$\text{Var}(a) = 0$$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

$$\text{Var}(X+Y) = ?$$

$$\mu_w = \mu_x + \mu_y$$

$$\text{Let } W = X + Y$$

$$\text{Var}(W) = E(W^2) - \mu_w^2$$

$$= E(X^2 + Y^2 + 2XY) - (\mu_x + \mu_y)^2$$

$$= E(X^2) + E(Y^2) + 2E(XY) - \mu_x^2 - \mu_y^2 - 2\mu_x\mu_y$$

$$= [E(X^2) - \mu_x^2] + [E(Y^2) - \mu_y^2] + 2[E(XY) - \mu_x\mu_y]$$

$$= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Cov}(X, Y) = E(XY) - \mu_x\mu_y$$

$$\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$$

$$\text{Corr}(X, Y) = \rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{Cov}(X, Y) = \rho_{X,Y} \sigma_X \sigma_Y$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\rho_{X,Y}\sigma_X\sigma_Y$$

$$\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$$

$$\begin{aligned}\text{Var}(aX+bY) &= a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y) \\ &= a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\rho_{X,Y}\sigma_X\sigma_Y\end{aligned}$$

You have \$1 to invest between two investments.

Let X be RV representing the return on \$1 in Inv 1

Let Y be RV representing the return on \$1 in Inv 2

$$E(X) = E(Y) > 0. \quad \text{Var}(X), \text{Var}(Y), \rho_{X,Y}$$

Let " a " be the amount invested in Inv 1.

Portfolio Return

$$aX + (1-a)Y$$

$$\begin{aligned}\text{It follows that } E(\text{Portfolio Return}) &= E(aX + (1-a)Y) \\ &= E(X) = E(Y).\end{aligned}$$

Var(Portfolio Return) -

$$\text{Var}(aX + (1-a)Y) = a^2 \text{Var}(X) + (1-a)^2 \text{Var}(Y) + 2a(1-a) \rho_{X,Y} \sigma_X \sigma_Y$$